NAWAL GUERD

DevOps Engineer

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SKILLS & TOOLS

DevOps & Platform

Kubernetes | kubeadm | Docker | Ansible | Argo CD | GitOps | GitLab CI/CD | HashiCorp Vault

Networking & Security

Cilium | Istio | Service Mesh | Zero Trust | Network Policies | DMZ | Air-Gapped Envs | MetalLB | Kong | Nginx

Observability & Data

Prometheus | Grafana | Loki | K6 | PostgreSQL | Redis | Kafka / Strimzi | ClickHouse

Systems & Scripting

Linux | Windows Server | Active Directory | Python | Shell Scripting | Node.js

Security Background

Nmap | Wireshark | Snort | OpenVAS | OWASP ZAP

Soft Skills

Analytical Thinking | Team Collaboration | Problem Solving | Leadership

LANGUAGES

PROFILE

DevOps engineer operating production Kubernetes platforms end to end: GitOps delivery, infrastructure automation, and Zero Trust network security, including in air-gapped, multi-datacenter environments. Background in a Computer Systems Security master's, with a published IEEE paper on Zero Trust microsegmentation for cloud-native networks — I don't just deploy a cluster, I design who is allowed to talk to it.

PROFESSIONAL EXPERIENCE

DevOps Engineer ATM Mobilis

11/2024 – Present | Algiers, Algeria

- Deploy and operate production **Kubernetes** clusters with **kubeadm**, including in **air-gapped** environments with controlled connectivity through DMZ and gateway components.
- Design **GitOps** workflows with **Argo CD** and **GitLab CI/CD**, and automate provisioning and configuration with **Ansible**.
- Enforce **Zero Trust** network policies and segmentation with **Cilium**; contribute to **multi-datacenter** Kubernetes architectures with high-availability requirements.
- Expose and load-balance application traffic via **MetalLB, Kong and Nginx**; run the **Prometheus / Grafana / Loki** observability stack.
- Administer distributed platform components (**Kafka/Strimzi, PostgreSQL, Redis, HashiCorp Vault**) and run load/performance testing with **K6**.
- Work daily across Development, Security, Network and Infrastructure teams on platform engineering, troubleshooting and documentation.

Cybersecurity Bootcamp Cybertalents

07/2022 – 10/2022 | Remote

- Trained across network security, penetration testing and vulnerability assessment with industry-standard tooling.

Summer Research Internship Sonatrach

07/2022 | Algiers, Algeria

- Assessed IT infrastructure and proposed an open-source, containerized video surveillance solution for enterprise deployment.

Summer Internship ManyDesigns Afrique

07/2020 – 08/2020 | Algiers, Algeria

- Deployed enterprise web server infrastructure and designed a local network architecture with its access control policies.

EDUCATION

Master's Degree, Computer Systems Security USTHB

Arabic	Native
French	Fluent
English	Proficient

CERTIFICATIONS

Certified Kubernetes Administrator *(course)*
KodeKloud — 08/2025

GitLab CI Fundamentals
GitLab — 11/2024

Next-Generation Cyber Security
Huawei — 08/2024

Cybersecurity Essentials
CISCO — 08/2023

Fundamentals of Deep Learning
NVIDIA — 12/2022

INTERESTS

- CTF Competitions
- Open Source
- Technical Writing
- Research Conferences

2022 – 2024 Algiers, Algeria

Thesis: *Design and Implementation of a Zero Trust Security Solution for 5G Networks* — built an automated 5G deployment/cluster-management application and implemented Zero Trust policies with Istio, leading to the IEEE WCNC 2025 publication below.

Bachelor's Degree, Mathematics & Computer Science USTHB

2019 – 2022 Algiers, Algeria

Telecommunications & Networks specialization — class rank 2/69. Capstone: containerized a video-surveillance SOA app with Docker Compose and a full Prometheus/Grafana monitoring stack.

RESEARCH

A Multi-Layered Zero Trust Microsegmentation Solution for Cloud-Native 5G & Beyond Networks IEEE WCNC 2025

03/2025 italy

*C. Benzaid, **N. Guerd**, N. E. H. Rehouma, K. Zeraoulia, T. Taleb*

- Proposed **THAALOUB**, a Zero Trust framework for 3GPP-compliant, end-to-end security in cloud-native 5G/B5G networks, combining **service mesh** and **CNI** for multi-layered microsegmentation.
- Validated experimentally: improved security posture with minimal added latency or resource overhead.

Read the paper (PDF) →

SELECTED PROJECTS

- **DDoS Detection & Mitigation in SDN** *(2024)* — Mininet/Ryu-based SDN with a Random Forest classifier for real-time DDoS detection and automated mitigation.
- **Microservices Security on Azure** *(2024)* — Compared Zero Trust, perimeter-based and DevSecOps approaches; built a security framework for a microservices deployment on Azure.
- **Black-Box Penetration Test — USTHB** *(2024)* — Led a simulated security audit of the Faculty of Computer Science, from engagement contract to full audit report.
- **Intrusion Detection with Snort** *(2024)* — Simulated an attack on Windows Server 2012 R2 and tuned Snort with custom alert rules to detect it.
- **Enterprise VPN Deployment** *(2024)* — Three-tier VPN architecture (client, Active Directory, gateway) with authenticated, encrypted remote access.

LEADERSHIP

IT Team Leader Openminds Club, USTHB

2020 – 2023 Algiers, Algeria

Led the open-source student club's technical activities: organized Install Party 10 and Hacktoberfest, and ran workshops on open-source tooling.